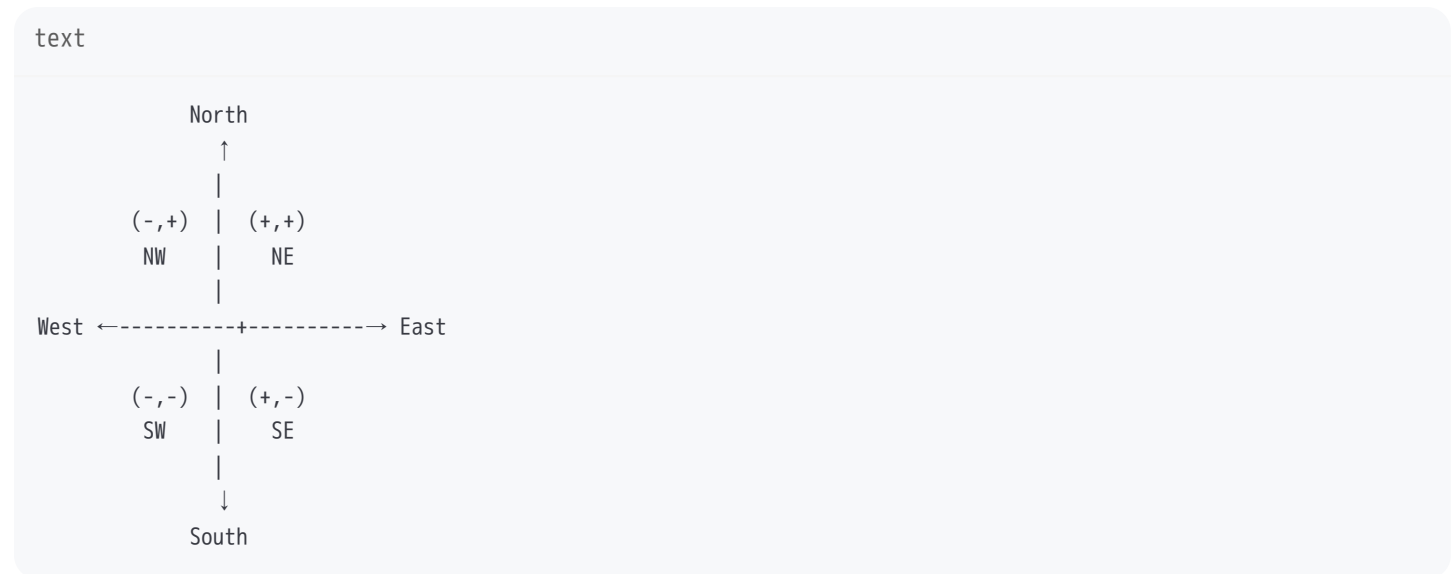


Brooo yes 🤔 I know the trick you mean. For **Direction Sense problems**, convert every movement into **X-Y coordinates**. Then at the end, just look at the signs. This makes even long statements easy.

🧭 Direction Sense — Coordinate Trick

1. Fix the coordinates

Think of yourself standing at **(0,0)**.



2. Remember only this

Direction	X	Y
East →	+	0
West ←	-	0
North ↑	0	+
South ↓	0	-

For diagonal directions:

Direction	X	Y
NE	+	+
NW	-	+
SE	+	-
SW	-	-

🔥 Main Trick

East/West changes X

North/South changes Y

That's it.

Suppose:

A walks 5 m East, then 3 m North.

Start:

text

$(0,0)$

East 5:

text

$(+5,0)$

North 3:

text

$(+5,+3)$

Both X and Y are positive:

text

$(+,+) = \text{NE}$

✔ Answer: **North-East**

Long Statements

Example:

A walks 10 m North, turns right and walks 5 m, then turns right and walks 7 m.

Don't worry about the turns initially. Convert the actual movement.

North:

text

$Y +10$

Right from North = East:

text

$X +5$

Right from East = South:

text

Y -7

Final:

text

X = +5

Y = +3

text

(+,+) → NE

✔ North-East

★ Most Important Final Sign Trick

After doing **ALL movements**, calculate:

text

X = East - West

Y = North - South

Then:

Final X	Final Y	Direction
+	+	NE
-	+	NW
+	-	SE
-	-	SW
+	0	East
-	0	West
0	+	North
0	-	South
0	0	Same position

Example

8 m East → 5 m North → 3 m West → 10 m South

Calculate:

text

X = +8 - 3 = +5

Y = +5 - 10 = -5

So:

text

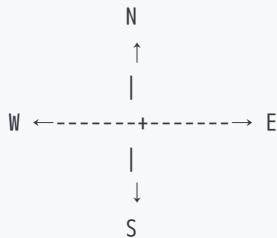
(+, -) = SE

✓ South-East

↻ Turning Trick

If questions say "turn right/left", remember:

text



Right turns

text

N → E
E → S
S → W
W → N

Left turns

text

N → W
W → S
S → E
E → N

Quick memory:

Right = clockwise ↻

Left = anticlockwise ↺

💡 One Important Point

If they ask "in which direction is A from B?"

Don't accidentally calculate B from A.

Example:

text

A = (5, 3)
B = (2, 7)

Direction of **A from B**:

text

A - B

X = 5 - 2 = +3

Y = 3 - 7 = -4

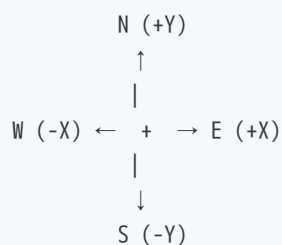
(+, -) = SE

✅ A is **South-East of B**.

Super Short Revision

Memorize this:

text



East = X+

West = X-

North = Y+

South = Y-

Then:

text

(+, +) NE

(-, +) NW

(+, -) SE

(-, -) SW

Do every movement → calculate final X,Y → check signs → answer. 🔥

 ChatGPT gpt-5-6

Yes bro 🔥 — after finding the **final coordinates**, you can calculate the **straight-line distance from the original starting point** using the Pythagoras formula.

Distance from starting point

If your final position is:

(X, Y)

and you started at **(0,0)**:

$$Distance = \sqrt{X^2 + Y^2}$$

Example

Suppose:

8 m East → 6 m North → 3 m West

Coordinates:

text

$$X = 8 - 3 = +5$$

$$Y = 6$$

Final position:

text

(5, 6)

Distance from starting point:

$$\begin{aligned} & \sqrt{5^2 + 6^2} \\ &= \sqrt{25 + 36} \\ &= \sqrt{61} \approx 7.81m \end{aligned}$$

✓ Distance = 7.81 m

Very important: Distance vs Direction

Direction: Look at the signs.

text

(+, +) → NE

(+, -) → SE

(-, +) → NW

(-, -) → SW

Distance: Ignore the signs and use:

$$\sqrt{X^2 + Y^2}$$

Easy shortcut

If final coordinates are:

text

(3, 4)

Distance = **5 m**

text

(5, 12) → 13 m

(6, 8) → 10 m

(8, 15) → 17 m

These are common **3-4-5 / 5-12-13 / 6-8-10** triangles, so recognize them quickly in aptitude questions.

One more thing: If the question asks **total distance travelled**, DON'T use Pythagoras. Add every movement. Pythagoras gives the **shortest straight-line distance from the starting point**.